

What Is Different — FC: UAVXArm32F4 → UAVXArmQ

Overview

UAVXArmQ is a quaternion-based fork of the original UAVXArm32F4 flight controller firmware. The attitude representation changed from Euler angles to quaternions internally, but the most visible differences from a user and parameter perspective are listed below.

Attitude Representation

Aspect	UAVXArm32F4 (old)	UAVXArmQ (new)
Internal attitude	Euler angles (roll, pitch, yaw)	Quaternion (q0, q1, q2, q3)
Gimbal lock risk	Yes (singularity at 90°pitch)	No
Computational cost	Lower (3 DOF)	Higher (4 DOF)
Sensor fusion	Complementary filter	Quaternion-based complementary filter

Impact: No gimbal lock. The aircraft can execute extreme manoeuvres (loops, flips) without the attitude estimate collapsing. The flight behaviour is otherwise identical from the pilot's perspective.

Parameter System — Unified Float

Old System (UAVXArm32F4 + Old GCS)

Parameters were a mix of `float32` and `uint8` depending on the parameter index: - Some parameters stored as 0-255 integers (e.g. gains, rates) - Other parameters stored as native floats (e.g. altitudes, voltages) - The GCS needed to know which format each parameter used - A **Legacy checkbox** in the old GCS toggled between `uint8` integer display and float display

New System (UAVXArmQ + UAVXGS)

All 128 parameters are `float32`: - Every parameter reads and writes as a decimal float over the wire - No `uint8` conversion, no `LEGACY_TAGS`, no `PARAM_SCALES` - The **Legacy checkbox is gone** — there is nothing to toggle

What This Means for the User

- **If you had a .af file from the old system:** The values that were `uint8` will now appear as equivalent floats (e.g. 150 → 150.0). The FC behaviour is identical — the FC internally casts `(float32)(uint8)value` on the old `uint8` params, so the round-trip is exact for 0-255.

- **If you used the Legacy checkbox:** You no longer need it. All params display as decimals. The conversion table that mapped uint8 ranges to display values has been removed — the display multiplier and limits now apply uniformly.
- **Parameter limits:** Previously, uint8 params had a hard limit of 0–255. Now all params use the full float32 range, limited per-index by `PARAM_LIMITS × PARAM_DISPLAY_MULT`.

Telemetry Changes

Aspect	UAVXArm32M	UAVXArmQ
Attitude control debug	Three separate per-axis packets (tag 57) at (tag 68) at 2.5 Hz	Single tuning packet at 50 Hz with all axes
Config / revision	Broadcast every other telemetry cycle (25 Hz)	Sent only on request (once on GCS connect)
Tuning data	Not available as a single frame	Rate tracking (de-sired/actual/out) for all axes in one 22-byte packet

Impact: Less wire bandwidth wasted on debug data that no one reads. The tuning data is now available in a single predictable packet that the GCS can plot or log.

Cruise Throttle Feedforward

Aspect	UAVXArm32M	UAVXArmQ
Tracking target	‘DesiredThrottle’ + AltHoldThrComp	‘AltHoldThrComp’ (direct)
Tracking rate	pCruiseThrFF‘ 0.5%/s	2%/s

Aspect	UAVXArm32	UAVXArmQ
ROC guard	Yes (could stall learning during gentle climbs)	No
Persistence	On disarm via 'figChanged'	Same, but not overridden by 'ApplyParameters()'
'pMaxAltHoldThrComp' default	15%	25%

Impact: Faster convergence to the correct hover throttle. The feedforward term is learned more aggressively and is no longer reset at parameter commit.

Arming

Aspect	UAVXArm32	UAVXArmQ
Arming channel	Usually Channel 5	Channel 6 (NavQualificationRC)
Switch arming	Physical arm switch	Physical arm switch on Ch6, AltHold always on, WPNav always on
Tx arming (pot)	Variable threshold	Arm >10%, AltHold >40% (in-flight), WPNav >60%
AltControlEnabled assignments	In 'MapRC()' only	Same — no change, confirmed

Impact: Arming is now consistently on Ch6 across both methods. The switch method enables all features unconditionally; the pot method provides progressive mode access.

Failsafe

Aspect	UAVXArm32M	UAVXArmQ
SBus failsafe detection	‘sbusDecode()’ sets ‘RC-Failsafe’	Same
Timeout detection	‘GenerateFailsafePacket()’ loads FS	Same
FS presets	NavQual=1650, Nav-Mode=RTH, sticks=neutral, throttle=idle	Same
Override at NavQual block	Forces AltHold + Angle-Mode	Same
PassThru behaviour	Overrides all failsafe	Same

Impact: No change. Failsafe behaviour is identical.

Emulation Constants (emu.h)

Aspect	UAVXArm32M	UAVXArmQ
Default hover throttle	Generic 0.55 (Ecks 2208+8×6, 800 g)	
Default mass	Generic 0.8 kg	
Heavy model	N/A EM_MODEL_HEAVY: 0.60 thr, 1.0 kg	

Impact: Emulation now matches a real airframe out of the box. The heavy variant matches the Racerstar 2205+5×3×3 build.